

Exercise Sheet 10 — Full Solutions

Solutions are written in English subparts (a), (b), (c), ...

1. Compute the following integrals

(a) $\int (2x + 3e^x - x\sqrt{x}) \, dx$ *Linearity and power rule.*

$$x^2 + 3e^x - \frac{2}{5}x^{5/2} + C$$

(b) $\int \left(5 \cos x + 3^x + \frac{1}{2\sqrt[3]{x}} \right) \, dx$ *Linearity; $\int a^x dx = \frac{a^x}{\ln a} + C$.* $5 \sin x + \frac{3^x}{\ln 3} + \frac{3}{4}x^{2/3} + C$

$\int \frac{(18 + 2x^2) \sin x + 5}{x^2 + 9} \, dx$ *Split: $(18+2x^2) \sin x$ ————— $x^2+9=2 \sin x$.* $-2 \cos x + \frac{5}{3} \arctan\left(\frac{x}{3}\right) + C$

$\int \frac{(\sqrt{x} + 2)^2}{\sqrt{x}} \, dx$ *Expand and use power rule.*

$$\frac{2}{3}x^{3/2} + 4x + 8\sqrt{x} + C$$

$\int \frac{3 - \cos^2 x}{\cos^2 x} \, dx$ *Rewrite as $\int \left(\frac{3}{\cos^2 x} - 1 \right) \, dx$.* $3 \tan x - x + C$

$\int \sqrt[3]{5 - 6x} \, dx$ Let $u = 5 - 6x$, $du = -6 \, dx$; then integrate $u^{1/3}$.

$$-\frac{1}{8} (5 - 6x)^{4/3} + C$$

$\int \frac{2x^3 + 8x - 5}{4 + x^2} \, dx$ *Polynomial division: $2x^3 + 8x - 5$ ————— $x^2+4=2x-\frac{5}{x^2+4}$.* $x^2 - \frac{5}{2} \arctan\left(\frac{x}{2}\right) + C$

$\int 3^{2-3x} \, dx$ Use $\int a^{kx} \, dx = \frac{a^{kx}}{k \ln a} + C$.

$$-\frac{3^{1-3x}}{\ln 3} + C$$

$\int \left(\frac{9}{9 + x^2} + \frac{5}{x^2} \right) \, dx$ *Linearity; $\int \frac{dx}{x^2} = -\frac{1}{x} + C$.* $3 \arctan\left(\frac{x}{3}\right) - \frac{5}{x} + C$

$\int \frac{dx}{1 + 2x^2}$ Use $\int \frac{dx}{1 + \alpha^2 x^2} = \frac{1}{\alpha} \arctan(\alpha x) + C$.

$$\frac{1}{\sqrt{2}} \arctan(\sqrt{2}x) + C$$

$\int \frac{dx}{\sqrt{36 - 3x^2}}$ Factor: $36 - 3x^2 = 3(12 - x^2)$ and use $\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin(x/a) + C$.

$$\frac{1}{\sqrt{3}} \arcsin\left(\frac{x}{2\sqrt{3}}\right) + C$$

2. Compute the following integrals

(a) $\int e^{-2x+1} dx$ Let $u = -2x + 1$.

$$-\frac{1}{2}e^{-2x+1} + C$$

(b) $\int \frac{dx}{1-3x}$ Let $u = 1 - 3x$.

$$-\frac{1}{3} \ln |1 - 3x| + C$$

(c) $\int \frac{\tan^2 x + 3x \sin^2 x}{\sin^2 x} dx$ Simplify to $\int (\sec^2 x + 3x) dx$.

$$\tan x + \frac{3}{2}x^2 + C$$

(d) $\int \frac{dx}{9-4x^2}$ Use the standard form $\int \frac{dx}{a^2-b^2x^2}$.

$$\frac{1}{12} \ln \left| \frac{3+2x}{3-2x} \right| + C$$

(e) $\int \frac{1+x+3x^2}{\sqrt{x}} dx$ Rewrite as $x^{-1/2} + x^{1/2} + 3x^{3/2}$.

$$2\sqrt{x} + \frac{2}{3}x^{3/2} + \frac{6}{5}x^{5/2} + C$$

(f) $\int \left(e^{5x} - \frac{2x^2+5}{1+x^2} \right) dx$ Split; write $\frac{2x^2+5}{1+x^2} = 2 + \frac{3}{1+x^2}$.

$$\frac{1}{5}e^{5x} - 2x - 3 \arctan x + C$$

(g) $\int \frac{dx}{(5-3x)^2}$ Let $u = 5 - 3x$.

$$\frac{1}{3(5-3x)} + C$$

(h) $\int \frac{dx}{3-16x^2}$ Use $\int \frac{dx}{a^2-b^2x^2}$ with $a = \sqrt{3}$, $b = 4$.

$$\frac{1}{8\sqrt{3}} \ln \left| \frac{\sqrt{3}+4x}{\sqrt{3}-4x} \right| + C$$

(i) $\int \frac{dx}{\sqrt{4-3x^2}}$ Use $\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin(x/a)$ after scaling.

$$\frac{1}{\sqrt{3}} \arcsin \left(\frac{\sqrt{3}x}{2} \right) + C$$

(j) $\int \frac{dx}{\sqrt{1 - (4x + 2)^2}}$ Let $u = 4x + 2$.

$$\frac{1}{4} \arcsin(4x + 2) + C$$

(k) $\int \frac{dx}{1 - 5x^2}$ Use $\int \frac{dx}{1 - \alpha^2 x^2} = \frac{1}{2\alpha} \ln \left| \frac{1 + \alpha x}{1 - \alpha x} \right| + C$.

$$\frac{1}{2\sqrt{5}} \ln \left| \frac{1 + \sqrt{5}x}{1 - \sqrt{5}x} \right| + C$$

3. Compute the following integrals (use substitution)

(a) $\int \frac{4x^3 - 1}{x^4 - x + 5} dx$ Let $u = x^4 - x + 5$, $du = (4x^3 - 1) dx$.

$$\ln |x^4 - x + 5| + C$$

(b) $\int \frac{\sqrt[4]{\ln x}}{x} dx$ Let $u = \ln x$, $du = \frac{dx}{x}$.

$$\frac{4}{5} (\ln x)^{5/4} + C$$

(c) $\int \frac{3x}{(x^2 + 4)^3} dx$ Let $u = x^2 + 4$, $du = 2x dx$.

$$-\frac{3}{4(x^2 + 4)^2} + C$$

(d) $\int \frac{e^{\tan x}}{\cos^2 x} dx$ Let $u = \tan x$, $du = \frac{dx}{\cos^2 x}$.

$$e^{\tan x} + C$$

(e) $\int \frac{\cos x}{\sqrt{\sin^2 x + 3}} dx$ Let $u = \sin x$, $du = \cos x dx$.

$$\ln \left| \sin x + \sqrt{\sin^2 x + 3} \right| + C$$

(f) $\int \frac{\sqrt{\arctan x}}{1 + x^2} dx$ Let $u = \arctan x$, $du = \frac{dx}{1 + x^2}$.

$$\frac{2}{3} (\arctan x)^{3/2} + C$$

(g) $\int \sqrt[3]{2 + 7 \sin x} \cos x dx$ Let $u = 2 + 7 \sin x$, $du = 7 \cos x dx$.

$$\frac{3}{28} (2 + 7 \sin x)^{4/3} + C$$

(h) $\int \frac{1}{x^2} \cos\left(\frac{1}{x}\right) dx$ Let $u = \frac{1}{x}$, $du = -\frac{1}{x^2} dx$.

$$\boxed{-\sin\left(\frac{1}{x}\right) + C}$$

(i) $\int \frac{\cos(5x)}{\sin^2(5x) + 1} dx$ Let $u = \sin(5x)$, $du = 5 \cos(5x) dx$.

$$\boxed{\frac{1}{5} \arctan(\sin(5x)) + C}$$

(j) $\int \frac{\ln x + 2}{2x} dx$ Let $u = \ln x$, $du = \frac{dx}{x}$.

$$\boxed{\frac{(\ln |x|)^2}{4} + \ln |x| + C}$$

(k) $\int \frac{dx}{\sin^2 x \sqrt{3 - \cot x}}$ Let $u = 3 - \cot x$, $du = \frac{dx}{\sin^2 x}$.

$$\boxed{2\sqrt{3 - \cot x} + C}$$

4. Compute the following integrals (integration by parts)

(a) $\int e^{-x}(x^2 + 5) dx$ Use repeated integration by parts for $x^2 e^{-x}$; then add $5 \int e^{-x} dx$.

$$\boxed{-e^{-x}(x^2 + 2x + 7) + C}$$

(b) $\int \arcsin x dx$ Take $u = \arcsin x$, $dv = dx$.

$$\boxed{x \arcsin x + \sqrt{1 - x^2} + C}$$

(c) $\int x \sin(3x) dx$ Take $u = x$, $dv = \sin(3x) dx$.

$$\boxed{-\frac{x \cos(3x)}{3} + \frac{\sin(3x)}{9} + C}$$

(d) $\int e^x \sin(3x) dx$ Use the standard two-step by-parts method (or complex exponentials).

$$\boxed{\frac{e^x}{10} (\sin(3x) - 3 \cos(3x)) + C}$$

(e) $\int x \cdot 10^x dx$ Take $u = x$, $dv = 10^x dx$ with $\int 10^x dx = \frac{10^x}{\ln 10}$.

$$\boxed{\frac{x 10^x}{\ln 10} - \frac{10^x}{(\ln 10)^2} + C}$$

(f) $\int (x-2) \ln(2x) dx$ Take $u = \ln(2x)$, $dv = (x-2) dx$.

$$\left(\frac{x^2}{2} - 2x \right) \ln(2x) - \frac{x^2}{4} + 2x + C$$

(g) $\int \arccos(2x) dx$ Take $u = \arccos(2x)$, $dv = dx$ and substitute $w = 1 - 4x^2$ in the remaining integral.

$$x \arccos(2x) - \frac{1}{2} \sqrt{1 - 4x^2} + C$$

(h) $\int \ln(x+2) dx$ Take $u = \ln(x+2)$, $dv = dx$.

$$(x+2) \ln(x+2) - (x+2) + C$$

(i) $\int \cos(\ln x) dx$ Substitute $u = \ln x$, so $dx = e^u du$, then integrate $\int e^u \cos u du$.

$$\frac{x}{2} (\cos(\ln x) + \sin(\ln x)) + C$$

(j) $\int x^2 \ln x dx$ Take $u = \ln x$, $dv = x^2 dx$.

$$\frac{x^3}{3} \ln x - \frac{x^3}{9} + C$$

(k) $\int x^2 \cos^2 x dx$ Use $\cos^2 x = \frac{1+\cos(2x)}{2}$ and integrate by parts for $\int x^2 \cos(2x) dx$.

$$\frac{x^3}{6} + \frac{x^2}{4} \sin(2x) + \frac{x}{4} \cos(2x) - \frac{1}{8} \sin(2x) + C$$

5. Compute the following partial-fractions-type integrals

(a) $\int \frac{10 dx}{15+4x}$ Let $u = 15+4x$.

$$\frac{5}{2} \ln |15+4x| + C$$

(b) $\int \frac{dx}{4-7x}$ Let $u = 4-7x$.

$$-\frac{1}{7} \ln |4-7x| + C$$

(c) $\int \frac{dx}{(x+15)^{20}}$ Power rule.

$$-\frac{1}{19(x+15)^{19}} + C$$

(d) $\int \frac{dx}{(5-2x)^8}$ Let $u = 5-2x$.

$$\frac{1}{14(5-2x)^7} + C$$

- (e) $\int \frac{x}{2x^2 + 2x + 5} dx$ Write $x = \frac{1}{4}(4x + 2) - \frac{1}{2}$; complete the square for the arctan term.

$$\boxed{\frac{1}{4} \ln(2x^2 + 2x + 5) - \frac{1}{6} \arctan\left(\frac{2x + 1}{3}\right) + C}$$

- (f) $\int \frac{x}{x^2 - 7x + 20} dx$ Write $x = \frac{1}{2}(2x - 7) + \frac{7}{2}$; complete the square for the arctan term.

$$\boxed{\frac{1}{2} \ln(x^2 - 7x + 20) + \frac{7}{\sqrt{31}} \arctan\left(\frac{2x - 7}{\sqrt{31}}\right) + C}$$

- (g) $\int \frac{x + 1}{5x^2 + 2x + 1} dx$ Split into a logarithm term plus an arctan term after completing the square.

$$\boxed{\frac{1}{10} \ln(5x^2 + 2x + 1) + \frac{2}{5} \arctan\left(\frac{5x + 1}{2}\right) + C}$$

- (h) $\int \frac{2x + 1}{x^2 - 4x + 5} dx$ Write $2x + 1 = (2x - 4) + 5$ and complete the square.

$$\boxed{\ln(x^2 - 4x + 5) + 5 \arctan(x - 2) + C}$$

- (i) $\int \frac{dx}{2x^2 - 2x + 3}$ Complete the square and use arctan.

$$\boxed{\frac{1}{\sqrt{5}} \arctan\left(\frac{2x - 1}{\sqrt{5}}\right) + C}$$

6. Compute the following rational-function integrals (use partial fractions)

- (a) $\int \frac{3x^3 + 5x^2 - 25x - 1}{(x + 2)(x - 1)^2} dx$ Partial fractions: $3 + \frac{5}{x+2} - \frac{6}{(x-1)^2}$.

$$\boxed{3x + 5 \ln|x + 2| + \frac{6}{x - 1} + C}$$

- (b) $\int \frac{x^2 - 3x + 17}{(x + 5)(x^2 + 2x + 4)} dx$ Decompose: $\frac{3}{x+5} + \frac{-2x+1}{x^2+2x+4}$.

$$\boxed{3 \ln|x + 5| - \ln(x^2 + 2x + 4) + \sqrt{3} \arctan\left(\frac{x + 1}{\sqrt{3}}\right) + C}$$

- (c) $\int \frac{x^2 + 3}{(x + 1)^2(x - 1)} dx$ Decompose: $\frac{1}{x-1} - \frac{2}{(x+1)^2}$.

$$\boxed{\frac{2}{x + 1} + \ln|x - 1| + C}$$

- (d) $\int \frac{x^3 + 2x + 5}{x^2 - 1} dx$ Decompose: $x - \frac{1}{x+1} + \frac{4}{x-1}$.

$$\boxed{\frac{x^2}{2} - \ln|x + 1| + 4 \ln|x - 1| + C}$$

(e) $\int \frac{x^3 + 3x^2 + 5x + 7}{x^2 + 2} dx$ Divide: $x + 3 + \frac{3x+1}{x^2+2}$.

$$\frac{x^2}{2} + 3x + \frac{3}{2} \ln(x^2 + 2) + \frac{1}{\sqrt{2}} \arctan\left(\frac{x}{\sqrt{2}}\right) + C$$

(f) $\int \frac{x^2 + 2x + 6}{(x-1)(x-2)(x-4)} dx$ Decompose: $\frac{3}{x-1} - \frac{7}{x-2} + \frac{5}{x-4}$.

$$3 \ln|x-1| - 7 \ln|x-2| + 5 \ln|x-4| + C$$

(g) $\int \frac{2x^2 + x + 3}{(x+2)(x^2+x+1)} dx$ Decompose: $\frac{3}{x+2} - \frac{x}{x^2+x+1}$.

$$3 \ln|x+2| - \frac{1}{2} \ln(x^2 + x + 1) + \frac{1}{\sqrt{3}} \arctan\left(\frac{2x+1}{\sqrt{3}}\right) + C$$

(h) $\int \frac{dx}{x^5 - x^2}$ Decompose: $-\frac{1}{x^2} + \frac{1}{3(x-1)} + \frac{-x+1}{3(x^2+x+1)}$.

$$\frac{1}{x} + \frac{1}{3} \ln|x-1| - \frac{1}{6} \ln(x^2 + x + 1) + \frac{\sqrt{3}}{3} \arctan\left(\frac{2x+1}{\sqrt{3}}\right) + C$$

7. Compute the following trigonometric integrals

(a) $\int \frac{\cos x}{1 + \cos x} dx$ Rewrite as $1 - \frac{1}{1+\cos x}$ and use $1 + \cos x = 2 \cos^2(x/2)$.

$$x - \tan\left(\frac{x}{2}\right) + C$$

(b) $\int \frac{dx}{4 \sin x + 3 \cos x + 5}$ Use the universal substitution $t = \tan(x/2)$ to get $\int \frac{dt}{(t+2)^2}$.

$$-\frac{1}{\tan(x/2) + 2} + C$$

(c) $\int \frac{dx}{5 \sin x + 3 \cos x + 3}$ Use $t = \tan(x/2)$ to get $\int \frac{dt}{3+5t}$.

$$\frac{1}{5} \ln \left| 3 + 5 \tan\left(\frac{x}{2}\right) \right| + C$$

(d) $\int \sin^5 x dx$ Write $\sin^5 x = (1 - \cos^2 x)^2 \sin x$ and set $u = \cos x$.

$$-\cos x + \frac{2}{3} \cos^3 x - \frac{1}{5} \cos^5 x + C$$

(e) $\int \sqrt[3]{\sin^2 x} \cos^3 x dx$ Write $\cos^3 x = (1 - \sin^2 x) \cos x$ and set $u = \sin x$.

$$\frac{3}{5} \sin^{5/3} x - \frac{3}{11} \sin^{11/3} x + C$$

(f) $\int \sin^2 x \cos^2 x \, dx$ Use $\sin^2 x \cos^2 x = \frac{1}{4} \sin^2(2x)$ and power reduction.

$$\boxed{\frac{x}{8} - \frac{\sin(4x)}{32} + C}$$

(g) $\int \sin^4(2x) \, dx$ Use $\sin^4 u = \frac{3-4\cos(2u)+\cos(4u)}{8}$ with $u = 2x$.

$$\boxed{\frac{3x}{8} - \frac{\sin(4x)}{8} + \frac{\sin(8x)}{64} + C}$$

(h) $\int \sin(5x) \cos x \, dx$ Use product-to-sum: $\sin A \cos B = \frac{1}{2}(\sin(A+B) + \sin(A-B))$.

$$\boxed{-\frac{\cos(6x)}{12} - \frac{\cos(4x)}{8} + C}$$

(i) $\int \cos(3x) \cos(2x) \, dx$ Use product-to-sum: $\cos A \cos B = \frac{1}{2}(\cos(A-B) + \cos(A+B))$.

$$\boxed{\frac{1}{2} \sin x + \frac{1}{10} \sin(5x) + C}$$

8. Compute the following integrals (irrational expressions)

(a) $\int \frac{x+1}{\sqrt[3]{3x+1}} \, dx$ Let $u = 3x+1$.

$$\boxed{\frac{1}{15}(3x+1)^{5/3} + \frac{1}{3}(3x+1)^{2/3} + C}$$

(b) $\int \frac{dx}{\sqrt[3]{(2x+1)^2} - \sqrt{2x+1}}$ Let $u = (2x+1)^{1/6}$ so $(2x+1) = u^6$.

$$\boxed{\frac{3}{2}u^2 + 3u + 3 \ln|u-1| + C, \quad u = (2x+1)^{1/6}}$$

(c) $\int \frac{\sqrt[6]{x}}{1+\sqrt[3]{x}} \, dx$ Let $u = x^{1/6}$ so $x = u^6$.

$$\boxed{\frac{6}{5}x^{5/6} - 2x^{1/2} + 6x^{1/6} - 6 \arctan(x^{1/6}) + C}$$

(d) $\int \frac{7x+16}{\sqrt[3]{(7x+8)^2} + 2\sqrt[3]{7x+8}} \, dx$ Let $u = (7x+8)^{1/3}$.

$$\boxed{\frac{3}{28}(7x+8)^{4/3} - \frac{2}{7}(7x+8) + \frac{6}{7}(7x+8)^{2/3} + C}$$